SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

MTH319

$\frac{20}{5} = 24$ min.

Examination - July Semester 2024

Mathematical Methods I

Friday, 15 November 2024 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **SIX (6)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a

Find all solutions to the system of linear equations

$$\begin{aligned} x + y + 2z &= 8 \\ -x - 2y + 3z &= 1 \\ 3x - 7y + 4z &= 10 \end{aligned}$$

by using Gauss-Jordan elimination. What type of solution(s) does it have and what does the solution(s) represent geometrically?

(12 marks)

Question 1b

If $a \neq 0$ and $ad - bc \neq 0$, show that the reduced row echelon form of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and this linear system

$$\begin{aligned} ax + by &= m \\ cx + dy &= n \end{aligned}$$

has exactly one solution.

(8 marks)

Question 2

Consider matrix $A = \begin{bmatrix} 4 & 1 \\ -1 & 6 \end{bmatrix}$.

Question 2a

Compute its eigenvalues and eigenvectors. What are the algebraic and geometric multiplicity of the eigenvalues?

(8 marks)

Question 2b

Hence, show that matrix A is defective.

(2 marks)

Question 2c

Determine the Jordan normal form of A.

(8 marks)

Question 2d

Hence or otherwise evaluate e^A.

(4 marks)

Question 3

The points $P(2,2,1)$ and $Q(6,-7,-1)$ lie on the plane Δ with Cartesian equation:

$$\alpha x + 4y - 12z = k$$

where α and k are constants. $\alpha = 3$ $k = 2$

Question 3a

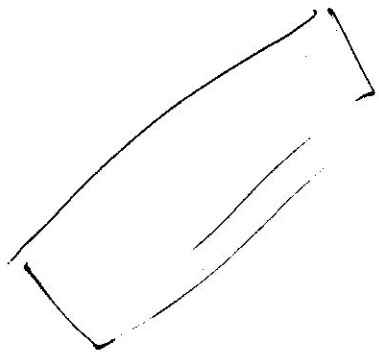
Determine the equation of the straight line L which is perpendicular to plane Δ and passing through point P.

(8 marks)

Question 3b

The points A and B are both located on L and each of these points is at a distance of 26 units from plane Δ. Show that the area of the triangle ABQ is approximately 261 square units.

(6 marks)



-28
$+2$
-16

Question 4

A 70 kg sky diver jumps out of a plane with no initial velocity. The velocity of the sky diver at any time t after jumping is $v(t)$ and the air resistance is $0.7|v|$. Here are the forces that are acting on the sky diver

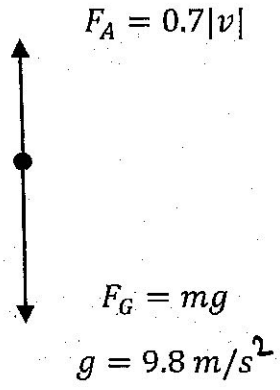


Figure Q4

Question 4a

Assume that the positive direction is upward, form an initial value problem and show that the rate of change of the diver's velocity is

$$\frac{dv}{dt} = -9.8 - \frac{v}{100}, v(0) = 0.$$

(6 marks)

Question 4b

What is the unit of $v(t)$?

(1 mark)

Question 4c

Determine the solutions of the initial value problem in Question 4a to find velocity as a function of time.

(10 marks)

Question 4d

Find the diver's velocity after 10s. Explain the negative velocity in your answer.

(2 marks)

Question 5a

Show that the following 2nd order linear equation: $y'' - 4y' + 5y = 0$, is equivalent to this system of 1st order linear equations:

$$y' = x$$
$$x' = -5y + 4x.$$

(4 marks)

Question 5b

Rewrite the system of 1st order linear equations in Question 5a into a matrix-vector equation $\overrightarrow{x'} = A\overrightarrow{x}$ where $\overrightarrow{x} = \begin{bmatrix} y \\ x \end{bmatrix}$. Find the eigenvalues and eigenvectors of matrix A.

(14 marks)

Question 5c

Hence, solve the system of linear differential equations.

(7 marks)

----- END OF PAPER -----